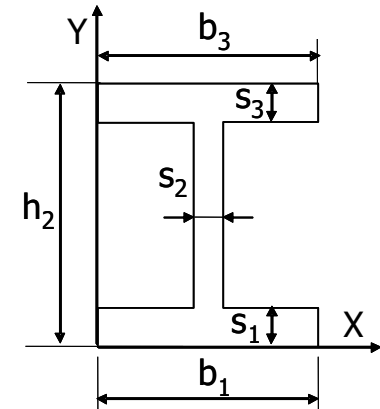


# Fundamentals of Strength of Materials – 2018-2019

## Tutorial #5: Cross sections and axial loading

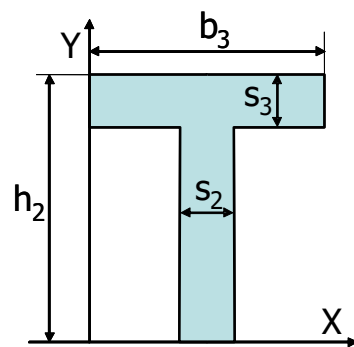
### Example 1

Find the position and the directions of the centroidal axes and compute the principal moments of area of second order for the W-shape and T-shape areas shown below.



$b_1 = 200 \text{ mm}$   
 $h_2 = 300 \text{ mm}$   
 $b_3 = 200 \text{ mm}$   
 $s_1 = 4 \text{ mm}$   
 $s_2 = 6 \text{ mm}$   
 $s_3 = 4 \text{ mm}$

$X_c = 100.0$ ;  $Y_c = 150.0 \text{ mm}$ ;  
 $I_{xx} = 4.75 \cdot 10^7$ ;  $I_{yy} = 5.34 \cdot 10^6$ ;  $I_{xy} = 0 \text{ mm}^4$ .



$h_2 = 300 \text{ mm}$   
 $b_3 = 200 \text{ mm}$   
 $s_2 = 6 \text{ mm}$   
 $s_3 = 8 \text{ mm}$

$X_c = 100.00$ ;  $Y_c = 217.60 \text{ mm}$ ;  
 $I_{xx} = 3.13 \cdot 10^7$ ;  $I_{yy} = 5.34 \cdot 10^6$ ;  $I_{xy} = 0 \text{ mm}^4$ .

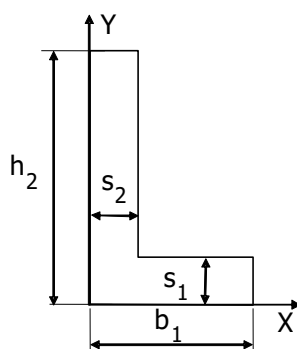
### Example 2

Find the centroidal axes and principal moments of area of second order for the following hollow cross section:

1. circular cross section, external diameter  $D = 400 \text{ mm}$  and internal diameter  $d = 300 \text{ mm}$ ;
2. rectangular cross section, external side  $B = 200$ ,  $H = 300 \text{ mm}$ ; internal side  $b = 150$ ,  $h = 250 \text{ mm}$ ;

### Example 3

For the L section shown, find the location of the centroidal axes, compute the principal moments of area of the second order and their orientation with respect to the xy coordinates shown in the figure.



$b_1 = 80 \text{ mm}$   
 $h_2 = 200 \text{ mm}$   
 $s_1 = 13 \text{ mm}$   
 $s_2 = 10 \text{ mm}$

Compute and plot the normal stresses on the cross section of beam, due to a normal load  $N = 250 \text{ kN}$  and the safety factor, considering that the beam is made of cast iron ( $S_u = 300 \text{ MPa}$ ;  $A = 2\%$ ).

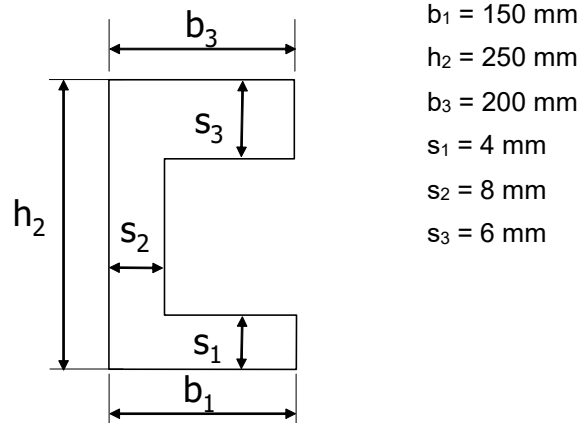
$X_c = 17.51$ ;  $Y_c = 70.76 \text{ mm}$ ;  $I_{xx} = 12.15 \cdot 10^6$ ;  $I_{yy} = 1.39 \cdot 10^6$ ;  $I_{xy} = -2.34 \cdot 10^6 \text{ mm}^4$ ;  
 $\alpha(x^{\wedge}p_1) = -11.75^\circ$ ;  $I_1 = 12.63 \cdot 10^6$ ;  $I_2 = 0.90 \cdot 10^6 \text{ mm}^4$ ;

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### Example 4

Find the position of the centroidal axes and the principal moments of area of the second order for the C-shape here depicted.



Compute and plot the related distribution of normal stresses on the cross section, corresponding to a normal load  $N = -440 \text{ kN}$  and compute the safety factor ( $S_y = 235 \text{ MPa}$ ;  $A = 15\%$ ) according to the Tresca failure criterion.

$$X_c = 46.42; Y_c = 144.00 \text{ mm}; I_{xx} = 34.82 \cdot 10^6; I_{yy} = 12.53 \cdot 10^6; I_{xy} = 5.82 \cdot 10^6 \text{ mm}^4;$$

$$\alpha(x^{\wedge}p_1) = 13.8^\circ; I_1 = 36.24 \cdot 10^6; I_2 = 11.10 \cdot 10^6 \text{ mm}^4;$$